

WASP-43b

R-1802

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220222 · created 2026-08-03T02:02:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459632.8443 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.179 +/- 0.018
Transit depth (Rp/Rs)^2	3.21 +/- 0.63 [%]
Orbital Inclination (inc)	81.38 +/- 1.7
Airmass coefficient 1 (a1)	1.137 +/- 0.016
Airmass coefficient 2 (a2)	-0.09 +/- 0.032
Scatter in the residuals of the lightcurve fit is	1.42 %
Best Comparison Star	#3 - [624, 344]
Optimal Method	PSF photometry
Transit Duration (day)	0.0494 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.179 ± 0.018 vs 0.1594 (deviation 1.09σ)
Duration fit vs published	0.0494 d vs 0.0512 d (deviation 0.29σ)
Mid-time O–C	-1.5 min
Depth SNR	9.9
Residual scatter	1.42 %

Light curve

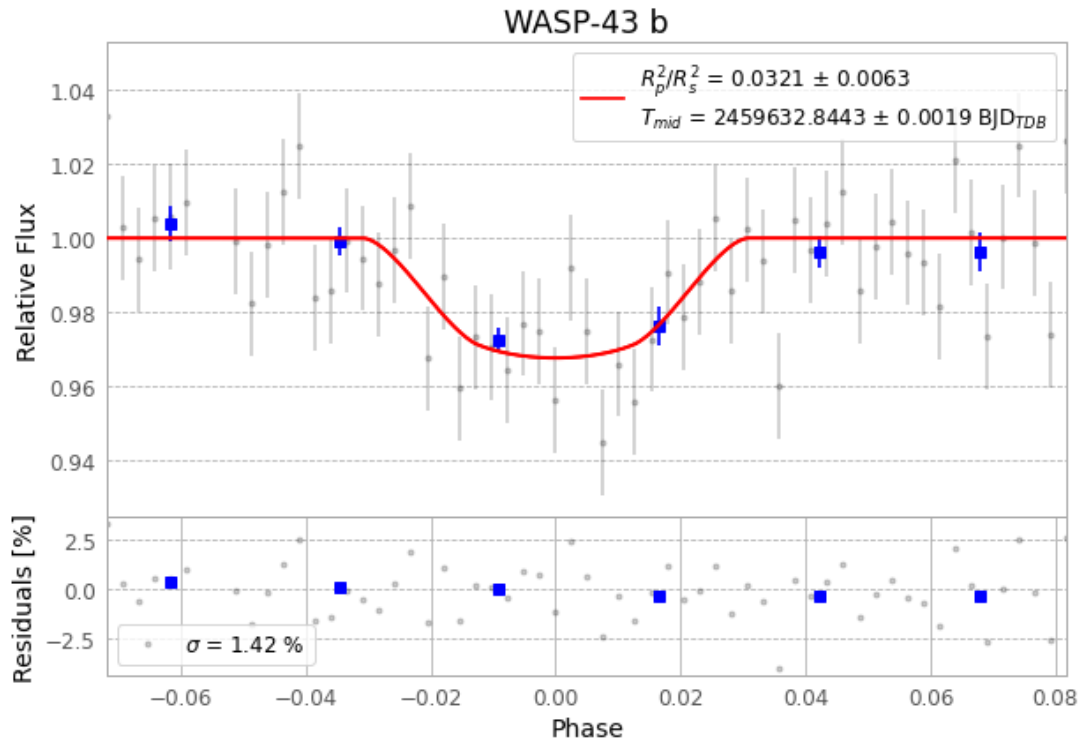


Figure 1 — FinalLightCurve_WASP-43 b_2022-02-21.png (SHA-256 28f250deb07d905f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 260], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a9fe001f5f074f4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

59 FITS frames (39 MB), WASP-43220222064212.FITS → WASP-43220222094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220222.log (7324cbd66cf2...), FinalLightCurve_WASP-43 b_2022-02-21.png (28f250deb07d...), FinalLightCurve_WASP-43 b_2022-02-21.csv (b338610c1101...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 02:02Z.